

The Super Learner

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Problem

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How do we find this function?

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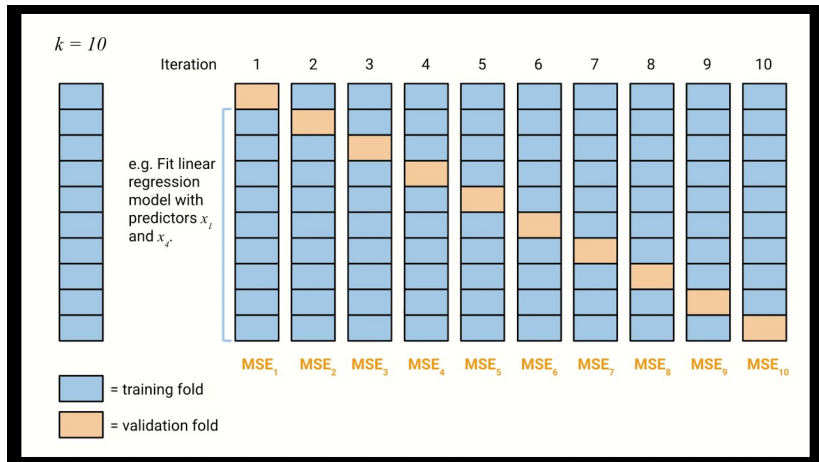
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$$\psi_0(W) = \arg \min_{\psi \in \mathcal{L}} E[L(X, \psi(W))],$$

where the loss function L is usually the squared loss $L = (Y - \psi(W))^2$. This is done by using cross validation.

10 fold crossvalidation



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 - Not as fancy as the ensemble learner

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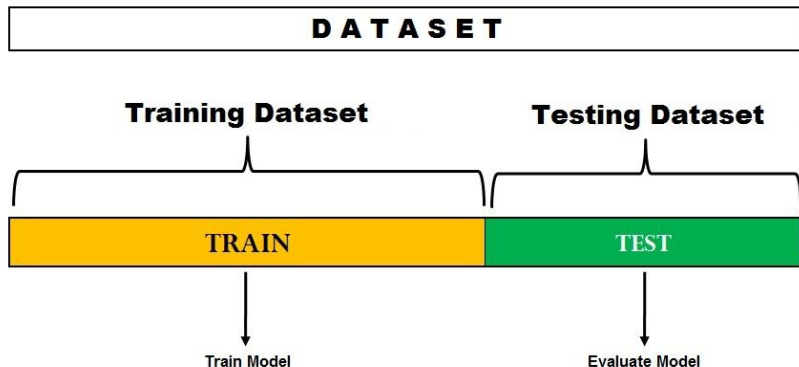
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 - Crazy bad runtime

Evaluate the model

Normally shown by discrimination- and calibration-plots, and measured by the AUC and the Brier-score, but on what data?

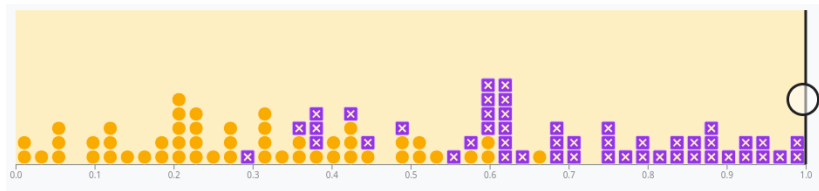
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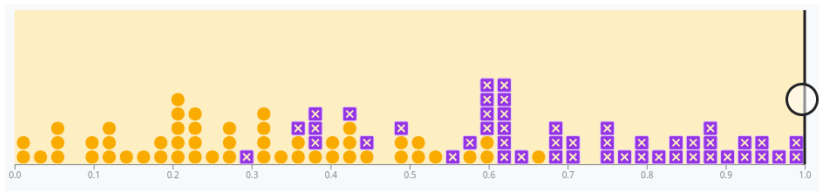
Performance measures: AUC

Measures discrimination, i.e. how good is the model to discriminate between a person who gets diabetes and a person who doesn't. The AUC for a model is the probability of the model, given a random person who gets T2D and a random person who doesn't, to give the T2D person a higher risk.



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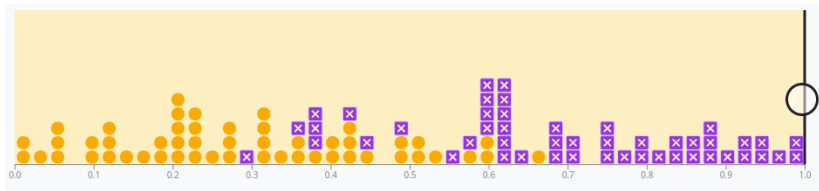
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Go to <https://developers.google.com/machine-learning/crash-course/classification/roc-and-auc> for an interactive display of the AUC

AUC plots

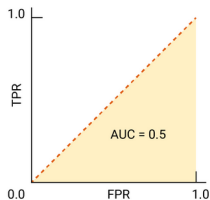


Figure 3. ROC and AUC of completely random guesses.

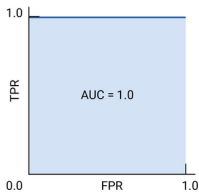


Figure 1. ROC and AUC of a hypothetical perfect model.

AUC plots

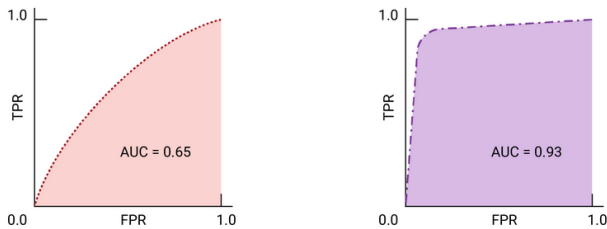
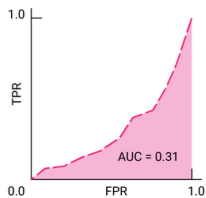


Figure 4. ROC and AUC of two hypothetical models. The curve on the right, with a greater AUC, represents the better of the two models.

AUC plots



Performance measures: Brier-score

Measure overall performance of a prediction model, and can be used as a standalone measure as it is a theoretical proper measure.

$$BS = \frac{1}{n} \sum_{i=1}^n (\psi_0(W_i) - Y_i)^2$$

Calibration plots

Calibration in the large refers to if the model average predicted probability is the same as the observed average probability of the outcome.

Calibration in the small refers to if the predicted probabilities match the observed probabilities, i.e. 25% of the people who get a prediction of 25% should get the event.